

Summary

Detail-oriented and results-driven Software Engineer with 3+ years of experience in designing and implementing test automation frameworks, backend development using Python and AWS Lambda, and ensuring software quality across CI/CD pipelines. Proven ability to collaborate with cross-functional teams to deliver scalable, secure, and high-performance applications. Adept at BDD-driven development, writing robust automated tests, and ensuring compliance with security frameworks such as GDPR and HIPAA.

Technical Skills:

Programming Languages: Python, Java, JavaScript (ES6+), SQL

Web Technologies: HTML5, CSS3, JavaScript, REST APIs

Automation & Testing: Selenium, Pytest, JUnit, BDD

CI/CD & DevOps: Jenkins, Azure DevOps, Bamboo, Git, Docker

Cloud & Serverless: AWS Lambda, EC2, S3, IAM

Databases: MySQL, PostgreSQL, MongoDB, Snowflake

Security Compliance: GDPR, HIPAA, SOC 2

Tools & Frameworks: Postman, JIRA, TestRail, GitHub Actions

Professional Experience:

Software Developer | Walmart – Remote, US | July 2023 – Present.

- Designed and implemented BDD test automation frameworks using Python (Behave) integrated into the CI/CD pipelines (Jenkins).
- Developed and maintained automated scripts for REST API and UI testing using Selenium and Pytest.
- Built and deployed serverless automation utilities using AWS Lambda for test result processing.
- Collaborated with developers and DevOps to ensure test environments were provisioned and integrated in Azure DevOps pipelines.
- Worked closely with product managers and stakeholders to define acceptance criteria and translate them into automated test cases.
- Performed security compliance checks aligned with HIPAA and GDPR standards for healthcare-related modules.
- Created dashboards for test coverage and defect analysis to monitor product quality over releases.
- Participated in regular code reviews ensuring adherence to test automation standards.
- Conducted root cause analysis and triaged test failures across test suites and environments.

Test Automation Engineer | TCS – India | Mar 2021 – Aug 2022.

- Developed automated test cases for data pipeline validations in data warehouse and OLAP systems.
- Wrote scalable and reusable Python scripts for backend and integration tests.
- Leveraged Postman and Pytest for API test automation and integrated it with Bamboo CI.
- Coordinated with developers to identify test coverage gaps and backfill automation suites.
- Used Cucumber with Java and Selenium to implement BDD-based regression suites.
- Designed data validation scripts for ETL workflows ensuring accuracy across PostgreSQL and Snowflake databases.
- Actively participated in defining test strategies aligned with sprint planning and agile methodologies.
- Documented detailed test plans, edge cases, and bug reports with reproduction steps in JIRA.
- Ensured secure handling of PII data in test environments to remain compliant with SOC 2 and GDPR.

Projects

Healthcare Portal Automation Suite

- Developed a BDD automation framework for a healthcare web app using Python + Behave + Selenium.
- Integrated test cases into Jenkins pipelines for nightly test runs and continuous validation.
- Simulated HIPAA-specific compliance test cases and ensured encryption validation of sensitive data.
- Created reusable page object models for UI automation across modules.
- Collaborated with dev and QA teams to triage failed test cases and integrate fixes.
- Generated reports using Allure for stakeholder visibility and test audit trails.

Serverless Test Orchestration Engine

- Built a Python-based automation trigger system using AWS Lambda + S3 + CloudWatch.
- Automated batch test execution for backend APIs and monitored execution through CloudWatch logs.
- Handled environment provisioning and test data setup via AWS SDK.
- Created custom scripts to parse test logs and notify via Slack and email.
- Enabled scheduling and on-demand test execution via API Gateway integrations.
- Reduced test setup time by 40% through infrastructure automation.

ETL Pipeline Validation Tool

- Designed an automated test utility to verify data integrity across ingestion, transformation, and export layers.
- Used SQL + Python + Pandas to validate source-to-target mappings in OLAP systems.
- Integrated with Snowflake and PostgreSQL to execute and validate dynamic queries.
- Included metadata validation, null checks, and referential integrity tests.
- Automated tool deployment into CI using Azure DevOps pipelines.
- Provided visualization of data mismatches and trend charts via Plotly reports.

Education:

Masters of Computer Science – University of Bridgeport, Connecticut | **Sep 2022 - May 2024**

CERTIFICATIONS

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- The Complete Web Development Bootcamp – Udemy
 - Full Stack Web Development with React Specialization – Coursera